

Appendix 3. Retail Process Load Baselines

Table 1a. Commercial kitchen appliance prescriptive measures and baseline for energy cost budget (IP units)

Appliance type	Baseline energy usage for energy modeling path				Levels for prescriptive path	
	Fuel	Function	Baseline efficiency	Baseline idle rate	Prescriptive efficiency	Prescriptive idle rate
Broiler, underfired	Gas	Cooking	30%	16,000 Btu/h/ft ² peak input	35%	12,000 Btu/h/ft ² peak input
Combination ovens, steam mode (P = pan capacity)	Elec	Cooking	40% steam mode	0.37P+4.5 kW	50% steam mode	0.133P+0.6400 kW
Combination ovens, steam mode	Gas	Cooking	20% steam mode	1,210P+35,810 Btu/h	38% steam mode	200P+6,511 Btu/h
Combination ovens, convection mode	Elec	Cooking	65% convection mode	0.1P+1.5 kW	70% convection mode	0.080P+0.4989 kW
Combination ovens, convection mode	Gas	Cooking	35% convection mode	322P+13,563 Btu/h	44% convection mode	150P+5,425 Btu/h
Convection oven, full-size	Elec	Cooking	65%	2.0 kW	71%	1.6 kW
Convection oven, full-size	Gas	Cooking	30%	18,000 Btu/h	46%	12,000 Btu/h
Convection oven, half-size	Elec	Cooking	65%	1.5 kW	71%	1.0 kW
Conveyor oven, > 25-inch belt	Gas	Cooking	20%	70,000 Btu/h	42%	57,000 Btu/h
Conveyor oven, ≤ 25-inch belt	Gas	Cooking	20%	45,000 Btu/h	42%	29,000 Btu/h
Fryer	Elec	Cooking	75%	1.05 kW	80%	1.0 kW
Fryer	Gas	Cooking	35%	14,000 Btu/h	50%	9,000 Btu/h
Griddle (based on 3 ft model)	Elec	Cooking	60%	400 W/ft ²	70%	320 W/ft ²

Griddle (based on 3 ft model)	Gas	Cooking	30%	3,500 Btu/h/ft ²	38%	2,650 Btu/h/ft ²
Hot food holding cabinets (excluding drawer warmers and heated display), 0 < V < 13 ft ³ (V = volume)	Elec	Cooking	na	40 W/ft ³	Na	21.5V Watts
Hot food holding cabinets (excluding drawer warmers and heated display), 13 ≤ V < 28 ft ³	Elec	Cooking	na	40 W/ft ³	Na	2.0V + 254 Watts
Hot food holding cabinets (excluding drawer warmers and heated display), 28 ft ³ ≤ V	Elec	Cooking	na	40 W/ft ³	Na	3.8V + 203.5 Watts
Large vat fryer	Elec	Cooking	75%	1.35 kW	80%	1.1 kW
Large vat fryer	Gas	Cooking	35%	20,000 Btu/h	50%	12,000 Btu/h
Rack oven, double	Gas	Cooking	30%	65,000 Btu/h	50%	35,000 Btu/h
Rack oven, single	Gas	Cooking	30%	43,000 Btu/h	50%	29,000 Btu/h
Range	Elec	Cooking	70%		80%	
Range	Gas	Cooking	35%	na	40% and no standing pilots	na
Steam cooker, batch cooking	Elec	Cooking	26%	200 W/pan	50%	135 W/pan
Steam cooker,	Gas	Cooking	15%	2,500 Btu/h/pan	38%	2,100 Btu/h/pan

batch cooking						
Steam cooker, high production or cook to order	Elec	Cooking	26%	330 W/pan	50%	275 W/pan
Steam cooker, high production or cook to order	Gas	Cooking	15%	5,000 Btu/h/pan	38%	4,300 Btu/h/pan
Toaster	Elec	Cooking	—	1.8 kW average operating energy rate	Na	1.2 kW average operating energy rate
Ice machine, IMH (ice-making head, $H =$ ice harvest), $H \geq 450$ lb./day	Elec	Ice	6.89 - 0.0011H kWh/100 lb. ice	na	$37.72 \cdot H^{-0.298}$ kWh/100 lb. ice	na
Ice machine, IMH (ice-making head), $H \leq 450$ lb./day	Elec	Ice	10.26 - 0.0086H kWh/100 lb. ice	na	$37.72 \cdot H^{-0.298}$ kWh/100 lb. ice	na
Ice machine, RCU (remote condensing unit, w/o remote compressor, $H < 1,000$ lb./day	Elec	Ice	8.85 - 0.0038H kWh/100lb ice	na	$22.95 \cdot H^{-0.258} + 1.00$ kWh/100 lb. ice	na
Ice machine, RCU (remote condensing unit), 1600 > $H \geq 1000$ lb./day	Elec	ice	5.10 kWh/100 lb. ice	Na	$22.95 \cdot H^{-0.258} + 1.00$ kWh/100 lb. ice	na
Ice machine, RCU (remote condensing unit), $H \geq 1600$ lb./day	Elec	Ice	5.10 kWh/100lb ice	Na	$-0.00011 \cdot H + 4.60$ kWh/100 lb. ice	na
Ice machine, SCU (self-	Elec	Ice	18.0 - 0.0469H	Na	$48.66 \cdot H^{-0.326} + 0.08$	na

contained unit), $H < 175$ lb./day			kWh/100lb ice		kWh/100 lb. ice	
Ice machine self-contained unit, $H \geq 175$ lb./day	Elec	Ice	9.80 kWh/100 lb. ice	Na	$48.66 \cdot H^{-0.326} + 0.08$ kWh/100 lb. ice	na
Ice machine, water-cooled ice-making head, $H \geq 1436$ lb./day (must be on chilled loop)	Elec	Ice	4.0 kWh/100 lb. ice	Na	3.68 kWh/100 lb. ice	na
Ice machine, water-cooled ice-making head, $500 \text{ lb./day} < H < 1436$ (must be on chilled loop)	Elec	Ice	$5.58 - 0.0011H$ kWh/100 lb. ice	Na	$5.13 - 0.001H$ kWh/100 lb. ice	na
Ice machine, water-cooled ice-making head, $H < 500$ lb./day (must be on chilled loop)	Elec	Ice	$7.80 - 0.0055H$ kWh/100 lb. ice	Na	$7.02 - 0.0049H$ kWh/100 lb. ice	na
Ice machine, water-cooled once-through (open loop)	Elec	Ice	Banned	Banned	Banned	Banned
Ice machine, water-cooled SCU (self-contained unit), $H < 200$ lb./day (must be on chilled loop)	Elec	Ice	$11.4 - 0.0190H$ kWh/100 lb. ice	Na	$10.6 - 0.177H$ kWh/100 lb. ice	na
Ice machine, water-cooled self-	Elec	Ice	7.6 kWh/100 lb. ice	Na	7.07 kWh/100 lb. ice	na

contained unit, $H \geq 200$ lb./day (must be on chilled loop)						
Chest freezer, solid or glass door	Elec	Refrig	$0.45V + 0.943$ kWh/day	Na	$\leq 0.270V + 0.130$ kWh/day	na
Chest refrigerator, solid or glass door	Elec	Refrig	$0.1V + 2.04$ kWh/day	Na	$\leq 0.125V + 0.475$ kWh/day	na
Glass-door reach-in freezer, $0 < V < 15$ ft ³	Elec	Refrig	$0.75V + 4.10$ kWh/day	Na	$\leq 0.607V + 0.893$ kWh/day	na
Glass-door reach-in freezer, $15 \leq V < 30$ ft ³	Elec	Refrig	$.75V + 4.10$ kWh/day	Na	$\leq 0.733V - 1.00$ kWh/day	na
Glass-door reach-in freezer, $30 \leq V < 50$ ft ³	Elec	Refrig	$.75V + 4.10$ kWh/day	Na	$\leq 0.250V + 13.50$ kWh/day	na
Glass-door reach-in freezer, $50 \leq V$ ft ³	Elec	Refrig	$0.75V + 4.10$ kWh/day	Na	$\leq 0.450V + 3.50$ kWh/day	na
Glass-door reach-in refrigerator, $0 < V < 15$ ft ³	Elec	Refrig	$0.12V + 3.34$ kWh/day	Na	$\leq 0.118V + 1.382$ kWh/day	na
Glass-door reach-in refrigerator, $15 \leq V < 30$ ft ³	Elec	Refrig	$0.12V + 3.34$ kWh/day	Na	$\leq 0.140V + 1.050$ kWh/day	na
Glass-door reach-in refrigerator, $30 \leq V < 50$ ft ³	Elec	Refrig	$0.12V + 3.34$ kWh/day	Na	$\leq 0.088V + 2.625$ kWh/day	na
Glass-door reach-in refrigerator, $50 \leq V$ ft ³	Elec	Refrig	$0.12V + 3.34$ kWh/day	Na	$\leq 0.110V + 1.500$ kWh/day	na

Solid-door reach-in freezer, $0 < V < 15 \text{ ft}^3$	Elec	Refrig	$0.4V + 1.38$ kWh/day	Na	$\leq 0.250V + 1.25$ kWh/day	na
Solid-door reach-in freezer, $15 \leq V < 30 \text{ ft}^3$	Elec	Refrig	$0.4V + 1.38$ kWh/day	Na	$\leq 0.400V - 1.000$ kWh/day	na
Solid-door reach-in freezer, $30 \leq V < 50 \text{ ft}^3$	Elec	Refrig	$0.4V + 1.38$ kWh/day	Na	$\leq 0.163V + 6.125$ kWh/day	na
Solid-door reach-in freezer, $50 \leq V \text{ ft}^3$	Elec	Refrig	$0.4V + 1.38$ kWh/day	Na	$\leq 0.158V + 6.333$ kWh/day	na
Solid-door reach-in refrigerator, $0 < V < 15 \text{ ft}^3$	Elec	Refrig	$0.1V + 2.04$ kWh/day	Na	$\leq 0.089V + 1.411$ kWh/day	na
Solid-door reach-in refrigerator, $15 \leq V < 30 \text{ ft}^3$	Elec	Refrig	$0.1V + 2.04$ kWh/day	Na	$\leq 0.037V + 2.200$ kWh/day	na
Solid-door reach-in refrigerator, $30 \leq V < 50 \text{ ft}^3$	Elec	Refrig	$0.1V + 2.04$ kWh/day	Na	$\leq 0.056V + 1.635$ kWh/day	na
Solid-door reach-in refrigerator, $50 \leq V \text{ ft}^3$	Elec	Refrig	$0.1V + 2.04$ kWh/day	Na	$\leq 0.060V + 1.416$ kWh/day	na
Clothes washer	Gas	Sanitation	1.72 MEF	Na	2.00 MEF	na
Door-type dish machine, high temp	Elec	Sanitation	na	1.0 kW	Na	0.70 kW
Door-type dish machine, low temp	Elec	Sanitation	na	0.6 kW	Na	0.6 kW
Multitank rack conveyor dish machine, high temp	Elec	Sanitation	na	2.6 kW	Na	2.25 kW

Multitank rack conveyor dish machine, low temp	Elec	Sanitation	na	2.0 kW	Na	2.0 kW
Single-tank rack conveyor dish machine, high temp	Elec	Sanitation	na	2.0 kW	Na	1.5 kW
Single-tank rack conveyor dish machine, low temp	Elec	Sanitation	na	1.6 kW	Na	1.5 kW
Undercounter dish machine, high temp	Elec	Sanitation	na	0.9 kW	Na	0.5 kW
Undercounter dish machine, low temp	Elec	Sanitation	na	0.5 kW	Na	0.5 kW

The energy efficiency, idle energy rates, and water use requirements, where applicable, are based on the following test methods:

ASTM F1275 Standard Test Method for Performance of Griddles

ASTM F1361 Standard Test Method for Performance of Open Deep Fat Fryers

ASTM F1484 Standard Test Methods for Performance of Steam Cookers

ASTM F1496 Standard Test Method for Performance of Convection Ovens

ASTM F1521 Standard Test Methods for Performance of Range Tops

ASTM F1605 Standard Test Method for Performance of Double-Sided Griddles

ASTM F1639 Standard Test Method for Performance of Combination Ovens

ASTM F1695 Standard Test Method for Performance of Underfired Broilers

ASTM F1696 Standard Test Method for Energy Performance of Single-Rack Hot Water Sanitizing, ASTM Door-Type Commercial Dishwashing Machines

ASTM F1704 Standard Test Method for Capture and Containment Performance of Commercial Kitchen Exhaust Ventilation Systems

ASTM F1817 Standard Test Method for Performance of Conveyor Ovens

ASTM F1920 Standard Test Method for Energy Performance of Rack Conveyor, Hot Water Sanitizing, Commercial Dishwashing Machines

ASTM F2093 Standard Test Method for Performance of Rack Ovens

ASTM F2140 Standard Test Method for Performance of Hot Food Holding Cabinets

ASTM F2144 Standard Test Method for Performance of Large Open Vat Fryers

ASTM F2324 Standard Test Method for Prerinse Spray Valves

ASTM F2380 Standard Test Method for Performance of Conveyor Toasters

ARI 810-2007: Performance Rating of Automatic Commercial Ice Makers

ANSI/ASHRAE Standard 72-2005: Method of Testing Commercial Refrigerators and Freezers with temperature setpoints at 38°F for medium-temp refrigerators, 0°F for low-temp freezers, and -15°F for ice cream freezers

Table 1b. Commercial Kitchen Appliance Prescriptive Measures and Baseline for Energy Cost Budget (SI units)

Appliance type	Baseline energy usage for energy modeling path				Levels for prescriptive path	
	Fuel	Function	Baseline efficiency	Baseline idle rate	Prescriptive efficiency	Prescriptive idle rate
Broiler, underfired	Gas	Cooking	30%	50.5 kW/m ²	35%	37.9 kW/m ²
Combination oven, steam mode (P = pan capacity)	Elec	Cooking	40% steam mode	0.37P+4.5 kW	50% steam mode	0.133P+0.6400 kW
Combination oven, steam mode	Gas	Cooking	20% steam mode	(1 210P+ 35 810)/3 412 kW	38% steam mode	(200P+6 511)/ 3 412 kW
Combination oven, convection mode	Elec	Cooking	65% convection mode	0.1P+1.5 kW	70% convection mode	0.080P+0.4989 kW
Combination oven, convection mode	Gas	Cooking	35% convection mode	(322P+ 13 563)/ 3412 kW	44% convection mode	(150P+5 425)/ 3412 kW
Convection oven, full-size	Elec	Cooking	65%	2.0 kW	71%	1.6 kW
Convection oven, full-size	Gas	Cooking	30%	5.3 kW	46%	3.5 kW
Convection oven, half-size	Elec	Cooking	65%	1.5 kW	71%	1.0 kW
Conveyor oven, > 63.5 cm belt	Gas	Cooking	20%	20.5 kW	42%	16.7 kW
Conveyor oven, < 63.5 cm belt	Gas	Cooking	20%	13.2 kW	42%	8.5 kW
Fryer	Elec	Cooking	75%	1.05 kW	80%	1.0 kW
Fryer	Gas	Cooking	35%	4.1 kW	50%	2.64 kW

Griddle (based on 90-cm model)	Elec	Cooking	60%	4.3 kW/m ²	70%	3.45 kW/m ²
Griddle (based on 90-cm model)	Gas	Cooking	30%	11 kW/m ²	33%	8.35 kW/m ²
Hot food holding cabinets (excluding drawer warmers and heated display) $0 < V < 0.368 \text{ m}^3$ (V = volume)	Elec	Cooking	na	1.4 kW/m ³	Na	$(21.5 \cdot V) / 0.0283$ kW/m ³
Hot food holding cabinets (excluding drawer warmers and heated display), $0.368 \leq V < 0.793 \text{ m}^3$	Elec	Cooking	na	1.4 kW/m ³	Na	$(2.0 \cdot V + 254) / 0.0283$ kW/m ³
Hot food holding cabinets (excluding drawer warmers and heated display), $0.793 \text{ m}^3 \leq V$	Elec	Cooking	na	1.4 kW/m ³	Na	$(3.8 \cdot V + 203.5) / 0.0283$ kW/m ³
Large vat fryer	Elec	Cooking	75%	1.35 kW	80%	1.1 kW
Large vat fryer	Gas	Cooking	35%	5.86 kW	50%	3.5 kW
Rack oven, double	Gas	Cooking	30%	19 kW	50%	10.25 kW
Rack oven, single	Gas	Cooking	30%	12.6 kW	50%	8.5 kW
Range	Elec	Cooking	70%	na	80%	na
Range	Gas	Cooking	35%	na	40% and no standing pilots	na
Steam cooker,	Elec	Cooking	26%	200 W/pan	50%	135 W/pan

batch cooking						
Steam cooker, batch cooking	Gas	Cooking	15%	733 W/pan	38%	615 W/pan
Steam cooker, high production or cook to order	Elec	Cooking	26%	330 W/pan	50%	275 W/pan
Steam cooker, high production or cook to order	Gas	Cooking	15%	1.47 kW/pan	38%	1.26 kW/pan
Toaster	Elec	Cooking	na	1.8 kW average operating energy rate	Na	1.2 kW average operating energy rate
Ice machine, IMH (ice making head, H = ice harvest) $H \geq 204$ kg/day	Elec	Ice	$0.0015 - 5.3464E^{-07}$ kWh/kg ice	na	$\leq 13.52 \cdot H^{-0.298}$ kWh/100 kg ice	na
Ice machine, IMH (ice making head), $H < 204$ kg/day	Elec	Ice	$0.2262 - 4.18E^{-04}$ kWh/kg ice	na	$\leq 13.52 \cdot H^{-0.298}$ kWh/100 kg ice	na
Ice machine, RCU (remote condensing unit, w/o remote compressor) $H < 454$ kg/day	Elec	Ice	$0.1951 - 1.85E^{-04}$ kWh/kg ice	na	$\leq 111.5835H^{-0.258} + 2.205$ kWh/100 kg ice	na
Ice machine, RCU (remote condensing unit) $726 > H \geq 454$ kg/day	Elec	Ice	0.1124 kWh/kg ice	na	$\leq 111.5835H^{-0.258} + 2.205$ kWh/100 kg ice	na
Ice machine, RCU (remote condensing unit), $H \geq 726$ kg/day	Elec	Ice	0.1124 kWh/kg ice	na	$\leq -0.00024H + 4.60$ kWh/100 kg ice	na

Ice machine, SCU (self-contained unit), $H < 79$ kg/day	Elec	Ice	$0.3968 - 2.28E^{-03}$ kWh/kg ice	na	$236.59H^{-0.326} + 0.176$ kWh/100 kg ice	na
Ice machine, SCU (self-contained unit), $H \geq 79$ kg/day	Elec	Ice	0.2161 kWh/kg ice	na	$236.59H^{-0.326} + 0.176$ kWh/100 kg ice	na
Ice machine, water-cooled ice-making head, $H \geq 651$ kg/day (must be on a chilled loop)	Elec	Ice	0.0882 kWh/kg ice	na	≤ 8.11 kWh/100 kg ice	na
Ice machine, water-cooled ice-making head, $227 \leq H < 651$ kg/day (must be on a chilled loop)	Elec	Ice	$0.1230 - 5.35E^{-05}$ kWh/kg ice	na	$\leq 11.31 - 0.065H$ kWh/100 kg ice	na
Ice machine, water-cooled ice-making head, $H < 227$ kg/day (must be on a chilled loop)	Elec	Ice	$0.1720 - 2.67E^{-04}$ kWh/kg ice	na	$\leq 15.48 - 0.0238H$ kWh/100 kg ice	na
Ice machine, water-cooled once-through (open loop)	Elec	Ice	Banned	Banned	Banned	Banned
Ice machine, water cooled SCU (self-contained unit) $H < 91$ kg/day (must be on a chilled loop)	Elec	Ice	$0.2513 - 9.23E^{-04}$ kWh/kg ice	na	$\leq 23.37 - 0.086H$ kWh/100 kg ice	na
Ice machine, water cooled SCU (self-contained unit) $H \geq 91$ kg/day (must	Elec	Ice	0.1676 kWh/kg ice	na	15.57 kWh/100 kg ice	na

be on a chilled loop)						
Chest freezer, solid or glass door	Elec	Refrig	15.90V + 0.943 kWh/day	na	9.541V + 0.130 kWh/day	na
Chest refrigerator, solid or glass door	Elec	Refrig	3.53V + 2.04 kWh/day	na	≤ 4.417 V + 0.475 kWh/day	na
Glass-door reach-in freezer, $0 < V < 0.42 \text{ m}^3$	Elec	Refrig	26.50V + 4.1 kWh/day	na	≤ 21.449V + 0.893 kWh/day	na
Glass-door reach-in freezer, $0.42 \leq V < 0.85 \text{ m}^3$	Elec	Refrig	26.50V + 4.1 kWh/day	na	≤ 25.901V - 1.00 kWh/day	na
Glass-door reach-in freezer, $0.85 \leq V < 1.42 \text{ m}^3$	Elec	Refrig	26.50V + 4.1 kWh/day	na	≤ 8.834V + 13.50 kWh/day	na
Glass-door reach-in freezer, $1.42 \leq V \text{ m}^3$	Elec	Refrig	26.50V + 4.1 kWh/day	na	≤ 15.90V + 3.50 kWh/day	na
Glass-door reach-in refrigerator, $0 < V < 0.42 \text{ m}^3$	Elec	Refrig	4.24V + 3.34 kWh/day	na	≤ 4.169V + 1.382 kWh/day	na
Glass-door reach-in refrigerator, $0.42 \leq V < 0.85 \text{ m}^3$	Elec	Refrig	4.24V + 3.34 kWh/day	na	≤ 4.947V + 1.050 kWh/day	na
Glass-door reach-in refrigerator, $0.85 \leq V < 1.42 \text{ m}^3$	Elec	Refrig	4.24V + 3.34 kWh/day	na	≤ 3.109V + 2.625 kWh/day	na
Glass-door reach-in refrigerator, $1.42 \leq V \text{ m}^3$	Elec	Refrig	4.24V + 3.34 kWh/day	na	≤ 3.887V + 1.500 kWh/day	na
Solid-door reach-in freezer, $0 < V < 0.42 \text{ m}^3$	Elec	Refrig	14.13V + 1.38 kWh/day	na	≤ 8.834V + 1.25 kWh/day	na

Solid-door reach-in freezer, $0.42 \leq V < 0.85 \text{ m}^3$	Elec	Refrig	14.13V + 1.38 kWh/day	na	$\leq 4.819V - 1.000$ kWh/day	na
Solid-door reach-in freezer, $0.85 \leq V < 1.42 \text{ m}^3$	Elec	Refrig	14.13V + 1.38 kWh/day	na	$\leq 5.760V + 6.125$ kWh/day	na
Solid-door reach-in freezer, $1.42 \leq V \text{ m}^3$	Elec	Refrig	14.13V + 1.38 kWh/day	na	$\leq 5.583V + 6.333$ kWh/day	na
Solid-door reach-in refrigerator, $0 < V < 0.42 \text{ m}^3$	Elec	Refrig	3.53V + 2.04 kWh/day	na	$\leq 3.145V + 1.411$ kWh/day	na
Solid-door reach-in refrigerator, $0.42 \leq V < 0.85 \text{ m}^3$	Elec	Refrig	3.53V + 2.04 kWh/day	na	$\leq 1.307V + 2.200$ kWh/day	na
Solid-door reach-in refrigerator, $0.85 \leq V < 1.42 \text{ m}^3$	Elec	Refrig	3.53V + 2.04 kWh/day	na	$\leq 1.979V + 1.635$ kWh/day	na
Solid-door reach-in refrigerator, $1.42 \leq V \text{ m}^3$	Elec	Refrig	3.53V + 2.04 kWh/day	na	$\leq 2.120V + 1.416$ kWh/day	na
Clothes washer	Gas	Sanitation	1.72 MEF		2.00 MEF	
Door-type dish machine, high temp	Elec	Sanitation	na	1.0 kW	Na	0.70 kW
Door-type dish machine, low temp	Elec	Sanitation	na	0.6 kW	Na	0.6 kW
Multitank rack conveyor dish machine, high temp	Elec	Sanitation	na	2.6 kW	Na	2.25 kW
Multitank rack conveyor dish machine, low temp	Elec	Sanitation	na	2.0 kW	Na	2.0 kW
Single-tank rack	Elec	Sanitation	na	2.0 kW	Na	1.5 kW

conveyor dish machine, high temp						
Single-tank rack conveyor dish machine, low temp	Elec	Sanitation	na	1.6 kW	Na	1.5 kW
Undercounter dish machine, high temp	Elec	Sanitation	na	0.9 kW	Na	0.5 kW
Undercounter dish machine, low temp	Elec	Sanitation	na	0.5 kW	Na	0.5 kW
The energy efficiency, idle energy rates, and water use requirements, where applicable, are based on the following test methods: ASTM F1275 Standard Test Method for Performance of Griddles ASTM F1361 Standard Test Method for Performance of Open Deep Fat Fryers ASTM F1484 Standard Test Methods for Performance of Steam Cookers ASTM F1496 Standard Test Method for Performance of Convection Ovens ASTM F1521 Standard Test Methods for Performance of Range Tops ASTM F1605 Standard Test Method for Performance of Double-Sided Griddles ASTM F1639 Standard Test Method for Performance of Combination Ovens ASTM F1695 Standard Test Method for Performance of Underfired Broilers ASTM F1696 Standard Test Method for Energy Performance of Single-Rack Hot Water Sanitizing, ASTM Door-Type Commercial Dishwashing Machines ASTM F1704 Standard Test Method for Capture and Containment Performance of Commercial Kitchen Exhaust Ventilation Systems ASTM F1817 Standard Test Method for Performance of Conveyor Ovens ASTM F1920 Standard Test Method for Energy Performance of Rack Conveyor, Hot Water Sanitizing, Commercial Dishwashing Machines ASTM F2093 Standard Test Method for Performance of Rack Ovens ASTM F2140 Standard Test Method for Performance of Hot Food Holding Cabinets ASTM F2144 Standard Test Method for Performance of Large Open Vat Fryers ASTM F2324 Standard Test Method for Prerinse Spray Valves ASTM F2380 Standard Test Method for Performance of Conveyor Toasters ARI 810-2007: Performance Rating of Automatic Commercial Ice Makers ANSI/ASHRAE Standard 72-2005: Method of Testing Commercial Refrigerators and Freezers with temperature setpoints at 38°F (3°C) for medium temp refrigerators, -18°C for low-temp freezers, and -26°C for ice cream freezers.						

Table 2. Supermarket refrigeration prescriptive measures and baseline for energy cost budget

Item	Attribute	Prescriptive measure	Baseline for energy modeling path
Commercial Refrigerator and Freezers	Energy Use Limits	ASHRAE 90.1-2010 Addendum g. Table 6.8.1L	ASHRAE 90.1-2010 Addendum g. Table 6.8.1L
Commercial Refrigeration Equipment	Energy Use Limits	ASHRAE 90.1-2010 Addendum g. Table 6.8.1M	ASHRAE 90.1-2010 Addendum g. Table 6.8.1M

Table 3. Walk-in coolers and freezers prescriptive measures and baseline for energy cost budget

Item	Attribute	Prescriptive measure	Baseline for energy modeling path
Envelope	Freezer insulation	R-46	R-36
	Cooler insulation	R-36	R-20
	Automatic closer doors	Yes	No
	High-efficiency low- or no-heat reach-in doors	40W/ft (130W/m) of door frame (low temperature), 17W/ft (55W/m) of door frame (medium temperature)	40W/ft (130W/m) of door frame (low temperature), 17W/ft (55W/m) of door frame (medium temperature)
Evaporator	Evaporator fan motor and control	Shaded pole and split phase motors prohibited; use PSC or EMC motors	Constant-speed fan
	Hot gas defrost	No electric defrosting.	Electric defrosting
Condenser	Air-cooled condenser fan motor and control	Shaded pole and split phase motors prohibited; use PSC or EMC motors; add condenser fan controllers	Cycling one-speed fan
	Air Cooled condenser design approach	Floating head pressure controls or ambient subcooling	10°F (-12°C) to 15°F (-9°C) dependent on suction temperature
Lighting	Lighting power density (W/sq.ft.)	0.6 W/sq.ft. (6.5 W/sq. meter)	0.6 W/sq.ft. (6.5 W/sq. meter)
Commercial Refrigerator and Freezers	Energy Use Limits	N/A	Use an Exceptional Calculation Method if attempting to take savings
Commercial Refrigerator and Freezers	Energy Use Limits	N/A	Use an Exceptional Calculation Method if attempting to take savings

Table 4. Commercial kitchen ventilation prescriptive measures and baseline for energy cost budget

Strategies	Prescriptive measure	Baseline
Kitchen hood control	ASHRAE 90.1-2010 Section 6.5.7.1, except that Section 6.5.7.1.3 and Section 6.5.7.1.4 shall apply if the total kitchen exhaust airflow rate exceeds 2,000 cfm (960 L/s) (as opposed to 5,000 cfm (2,400 L/s) noted in the ASHRAE 90.1-2010 requirements)	ASHRAE 90.1-2010 Section 6.5.7.1 and Section G3.1.1 Exception (d) where applicable